

AR Formula Card

Every formula, conversion, and word-problem pattern you need on test day. One page to glance at, one page to drill.

- 01 Order of operations + worked example
- 02 Fraction / decimal / percent table
- 03 Distance / Rate / Time triangle
- 04 Work-rate, mixture, percent change, ratios
- 05 Word-problem patterns ASVAB writers love
- 06 Test-day shortcuts

Order of operations (PEMDAS)

Parentheses, Exponents, Multiplication and Division (left to right), Addition and Subtraction (left to right). MD and AS share a tier — work left to right inside each.

$$P \rightarrow E \rightarrow M / D \rightarrow A / S$$

Worked: $6 + 4 \times (3 + 2)^2 \div 5 = 6 + 4 \times 5^2 \div 5 = 6 + 4 \times 25 \div 5 = 6 + 100 \div 5 = 6 + 20 = 26$

Fraction → decimal → percent

Memorize this table. Half of AR is faster if you know that 3/8 is .375 without thinking.

FRACTION	DECIMAL	PERCENT	FRACTION	DECIMAL	PERCENT
1/2	0.5	50%	3/5	0.6	60%
1/3	0.333	33.3%	4/5	0.8	80%
2/3	0.667	66.7%	1/6	0.167	16.7%
1/4	0.25	25%	1/8	0.125	12.5%
3/4	0.75	75%	3/8	0.375	37.5%
1/5	0.2	20%	5/8	0.625	62.5%
2/5	0.4	40%	1/10	0.1	10%

Distance / Rate / Time

D

R

T

$$D = R \times T \quad | \quad R = D / T \quad | \quad T = D / R$$

Cover the variable you need. The other two stay in the visible position. Always check that minutes and hours match before you multiply.

Ex 1: A convoy travels 240 miles in 4 hours. Rate? $R = D / T = 240 / 4 = 60$ mph.

Ex 2: A jet flies at 450 mph for 2.5 hours. Distance? $D = 450 \times 2.5 = 1,125$ miles.

Work-rate (combined work)

$$1/a + 1/b = 1/t$$

Add individual rates as fractions, then flip the sum to get total time. a and b = solo times.

Ex: Painter A finishes in 4 hr, Painter B in 6 hr. Together? $1/4 + 1/6 = 3/12 + 2/12 = 5/12$ per hour. Time = $12/5 = 2.4$ hr.

Mixture / average

$$\text{Average} = \text{Sum} / \text{Count} \quad | \quad \text{Weighted avg} = (q_1 v_1 + q_2 v_2) / (q_1 + q_2)$$

Mixture trick: if you mix 3 gal at 20% with 2 gal at 50%, total acid = $0.20(3) + 0.50(2) = 0.6 + 1.0 = 1.6$ gal in 5 gal → **32%**.

Percent change & the “1.X” trick

$$\% \text{ change} = (\text{New} - \text{Old}) / \text{Old} \times 100$$

Increase by X%: multiply by 1.X. Up 15%? $\times 1.15$. **Decrease by X%:** multiply by $(1 - 0.X)$. Down 20%? $\times 0.80$.

Ex: A \$80 jacket goes up 25%. New price = $80 \times 1.25 = \$100$. Then 30% off = $100 \times 0.70 = \$70$.

Ratio / proportion (cross-multiply)

$$a / b = c / d \rightarrow a \times d = b \times c$$

Ex: 3 MREs cost \$12. 7 MREs cost? $3/12 = 7/x \rightarrow 3x = 84 \rightarrow x = \28 .

Word-problem patterns ASVAB writers love

Recognize the cue, run the move. Don't solve the problem in your head — solve the pattern.

“How much faster / slower”

Think **rates**, not times. Convert both to mph (or units/hour) and subtract.

“Working together” / “at the same time”

Combined work-rate: $1/a + 1/b = 1/t$. Add the fractions, flip the sum.

“X% more” or “increased by X%”

Multiply the original by 1.X. Don't add then multiply again — it doubles up.

“What was the original price?”

Reverse percent. If sale price is 80% of original, divide sale price by 0.80.

“For every X, there are Y”

Set up a proportion. Cross-multiply. Don't reason it out in your head.

“On average” / “mean”

Sum / Count. For “what score do I need next time,” reverse it: Sum = Average $\times$ Count.

“Posts in a fence” / “floors in a building”

Off-by-one trap. A 100-ft fence with posts every 10 ft has **11 posts**, not 10.

“5 less than X” or “X fewer than Y”

Reverses order: 5 less than 12 = **12 – 5 = 7**, not 5 – 12. Write it out before translating.

Unit conversions worth memorizing

LENGTH		VOLUME		TIME / WEIGHT	
1 ft	12 in	1 gal	4 qt	1 hr	60 min
1 yd	3 ft	1 qt	2 pt	1 min	60 sec
1 mi	5,280 ft	1 pt	2 cups	1 lb	16 oz

Test-day shortcuts.

- **Estimate first.** Round to clean numbers. If your estimate is 150 and one choice is 1,500, you can eliminate it without finishing the math.
- **Eliminate before solving.** Cross off impossible answers (negative when problem requires positive, fraction when answer must be whole).
- **Plug in answer choices.** When algebra setup is slow, drop choice B or C into the problem. If it works, you're done. If too high, try A. Too low, try D.
- **Re-read the question before selecting.** “What was the tax?” vs “What was the total?” The wrong answer is usually sitting in the choices.
- **Never leave a CAT question blank.** Unanswered questions at the end of the CAT trigger a scoring penalty. A guess beats a blank.

The 4-step word-problem method

1. **Read the whole problem.** The final sentence tells you what to find.
2. **Underline what's asked.** "How many more" $\neq$ "how many total."
3. **Map buzzwords to operations.** "Per" = divide. "Of" = multiply. "Is" = equals.
4. **Build the equation, then solve.** Don't calculate until the setup is on paper.

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